

ONE HOT ENCODING

$$\overline{LMR} \quad \begin{matrix} 1 & 0 & 1 \\ S0 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$LMR \quad \begin{matrix} 0 & 0 & 1 \\ S1 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$\overline{LMR} \quad \begin{matrix} 0 & 1 & 1 \\ S2 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$LMR \quad \begin{matrix} 1 & 0 & 0 \\ S3 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$LMR \quad \begin{matrix} 1 & 1 & 0 \\ S4 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$\overline{LMR} \quad \begin{matrix} 1 & 1 & 1 \\ S5 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$LMR \quad \begin{matrix} 0 & 0 & 0 \\ S6 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

$$\overline{LMR} \quad \begin{matrix} 0 & 1 & 0 \\ S7 = \overline{LMR} \cdot (S0 + S1 + S2 + S3 + S4 + S5 + S6 + S7) \end{matrix}$$

~~Changes~~
→ add not gates to each input pin

Changes
→ swap
↔ NOT and
↔

$$DL = S0 + S1 + S2 + S3 + S4 + S5$$

$$DR = S0 + S1 + S2 + S3 + S4 + S5$$

← outputs

	<u>SR</u>	6	5	4	3	2	1	0
50%	S0 →	1	0	0	0	0	0	0
75%	S1 →	1	1	0	0	0	0	0
85%	S2 →	1	1	0	1	1	0	1
25%	S3 →	0	1	0	0	0	0	0
20%	S4 →	0	0	1	1	0	1	0
25%	S5 →	0	1	0	0	0	0	0
25%	S6 →	0	1	0	0	0	0	0
25%	S7 →	0	1	0	0	0	0	0

7 bits PWM

Control values

$$DC = \frac{CV}{2^n} \times 100$$

use less than output in comparator

	<u>SL</u>	6	5	4	3	2	1	0
50%	S0 →	1	0	0	0	0	0	0
25%	S1 →	0	1	0	0	0	0	0
20%	S2 →	0	0	1	1	0	1	0
75%	S3 →	1	1	0	0	0	0	0
85%	S4 →	1	1	0	1	1	0	1
50%	S5 →	1	0	0	0	0	0	0
50%	S6 →	1	0	0	0	0	0	0
50%	S7 →	1	0	0	0	0	0	0

$$\frac{50}{100} \times 2^7 = 64$$

$$\frac{25}{100} \times 2^7 = 32$$

$$\frac{75}{100} \times 2^7 = 96$$

$$\frac{20}{100} \times 2^7 = 25.6 \approx 26$$

$$\frac{85}{100} \times 2^7 = 108.8 \approx 109$$

SL

$$D6 = S0 + S3 + S4 + S5 + S6 + S7$$

$$D5 = S1 + S3 + S4$$

$$D4 = S2$$

$$D3 = S2 + S4$$

$$D2 = S4$$

$$D1 = S2$$

$$D0 = S4$$

To divide frequency until it is 3KHz

$$\frac{50 \times 10^6 \text{ Hz}}{128} = 390,625 \text{ Hz}$$

$$\frac{390,625 \text{ Hz}}{3,000 \text{ Hz}} = 130.20833$$

add FF

$$\frac{130.20833}{2} = 65.104165$$

MOD 66

$$256 - 66 = 190$$

1011 1110
MSB LSB

SR

$$D6 = S0 + S1 + S2$$

$$D5 = S1 + S2 + S3 + S5 + S6 + S7$$

$$D4 = S4$$

$$D3 = S2 + S4$$

$$D2 = S2$$

$$D1 = S4$$

$$D0 = S2$$